

Assignment 2

1. Write a program that prints numbers from 10 to 1 in descending order.

Example:

Output: 10 9 8 7 6 5 4 3 2 1

2. Write a program that reads 10 numbers from the user, calculates their average, and prints the result.

Example:

Input: 5 3 2 6 7 1 10 9 8 4

Output: 5.5

3. Write a program that displays the multiplication table of a given integer up to 12.

Example:

Input: 5

Output:

1 * 5 = 5

2 * 5 = 10

...

12 * 5 = 60

4. Write a program that allows the user to enter two integers, N and M, and prints the sum of all numbers between N and M (inclusive).

Note: -

if the user entered 5 and 8 then the output should be the sum of $5 + 6 + 7 + 8 = 26$

Example:

Input: 5 8

Output: 26

5. Write a program that allows the user to enter an indefinite number of integers until a negative number is entered. The program should then print the sum of all entered numbers (excluding the negative number).

Note: - when the negative number gets entered, the program must stop taking numbers from the user

Example:

Input: 5 7 8 3 2 -1

Output: 25

6. Write a program that allows the user to enter an indefinite number of integers until a negative number is entered. The program should then print the maximum value among the entered numbers.

Note: - when the negative number gets entered, the program must stop taking numbers from the user

Example:

Input: 5 3 8 7 9 4 -1

Output: 9

7. Write a program that takes an integer as input and counts the number of steps required to reduce it to zero using the following rules:
- If the number is odd, subtract 1.
 - If the number is even, divide it by 2.

Note: - here for example if the number is 15, then it must go through the following

$15 - 1 = 14$	(15 -> odd)	(step 1)
$14 / 2 = 7$	(14 -> even)	(step 2)
$7 - 1 = 6$	(7 -> odd)	(step 3)
$6 / 2 = 3$	(6 -> even)	(step 4)
$3 - 1 = 2$	(3 -> odd)	(step 5)
$2 / 2 = 1$	(2 -> even)	(step 6)
$1 - 1 = 0$	(1 -> odd)	(step 7)

Example:

Input: 15

Output: 7

8. Write a program to calculate the power of a number given a base and an exponent.

Example:

Input:

Base: 2

Exponent: 5

Output: $2^5 = 32$

9. Write a program that takes 10 numbers from the user and counts how many of them are odd.

Example:

Input: 5 8 3 11 6 4 2 7 1 9

Output: 6

10. Write a program that takes two integers from the user and finds their greatest common divisor (GCD).

Example:

Input: 25 15

Output: 5

11. Write a program that takes 10 numbers from the user and finds the second largest number among them.

Example:

Input: 5 8 3 11 6 4 2 7 1 9

Output: 9

Bonus Questions

1. Write a program to convert a decimal number to a binary number.

Note: - The Binary number must be printed as a whole number not a digit by digit

Example:

Input: 35

Output: 100011

2. Solve the following problems on Codeforces:

[Problem I]

(<https://codeforces.com/group/MWSDmqGsZm/contest/219432/problem/I>)

[Problem C]

(<https://codeforces.com/group/MWSDmqGsZm/contest/219432/problem/C>)